Eoin McLoughlin

Senior Machine Learning Engineer

eoin.mcloughlin1997@gmail.com | +353-89-952-2321 | eoinmcloughlin.com

GitHub | LinkedIn

Profile

Senior Machine Learning Engineer with 5+ years' experience building production ML systems, scalable data pipelines, and real-time inference platforms. Skilled in deep learning, time-series modelling, computer vision and LLM/RAG systems across cloud and edge environments. Experienced in translating research into production systems, leading ML infrastructure development and delivering models against real-world performance constraints.

Education

University College Dublin — *2025–Present*
MSc Advanced Artificial Intelligence

Dublin City University — *2016–2020*
BSc Computer Applications & Software Engineering — First Class Honours

Experience

MoveAhead (spinout from Dublin City University) — *Senior Machine Learning Engineer* — *2020 – 2025*
Tech: Python, PyTorch, TensorFlow, OpenCV, AWS (S3, EC2, Lambda), MongoDB, FastAPI, Docker, React, Android, C#

– Owned the end-to-end architecture and operation of the backend movement analysis platform, delivering production ML pipelines validated at >95% agreement with gold-standard motion capture.
– Established standards for pipeline development, feature engineering and model evaluation; led development of centralised feature store improving feature reuse and training/serving consistency.
– Architected ML data infrastructure and ingestion pipelines using S3, MongoDB and multiprocessing, reducing processing time by ∼ 80%.
– Built and maintained cross-platform applications (Android, React, C#) to deploy ML models on-device for millions of users.
– Technical lead for a 3-person backend ML team, defining engineering standards and guiding model development and evaluation.
– Designed, benchmarked and deployed object detection and pose estimation models for edge inference, achieving >30 FPS on constrained hardware.
– Designed and trained an LSTM-based sequence model for movement classification achieving 94% accuracy on real-world data.
– Partnered with founders and stakeholders to define KPIs, align ML development with product strategy, and translate requirements into deployable systems.

Projects

Agentic RAG Assistant for UCD — *2026*
Agentic RAG system for citation-backed academic policy, module and timetable queries
– Built a React/FastAPI assistant with ChromaDB, hybrid dense/BM25 retrieval and LLM function-calling.
– Created FAQ golden dataset and RAGAS/LLM-as-Judge evaluation, achieving 0.86 Pass@3 retrieval performance.

Financial Regime Classification (Classical vs Quantum ML) — *2025*
Market regime detection using ML and quantum kernel methods
– Built a time-series classification pipeline to identify market regimes using engineered financial features.
– Designed walk-forward, embargo-aware validation to prevent look-ahead bias in non-stationary data.
– Benchmarked XGBoost/LightGBM ensembles against quantum kernel SVMs (PennyLane).

Publications

Towards Efficient and Explainable HYDRA for Time Series Classification — McLoughlin, E. et al. — Submitted to the AALTD Workshop at ECML PKDD 2026.

Technical Skills

Languages: Python, Java, C#, SQL, JavaScript
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face
LLM & Agentic AI: RAG, LangChain, LangGraph, LoRA, LLM Function Calling, RAGAS
Data & Infrastructure: Spark, Hadoop, AWS (EC2, S3, Lambda), Docker, FastAPI, MongoDB
Techniques: Deep Learning, Computer Vision, Time Series Modelling, Feature Engineering, Model Optimisation
Focus: Production ML Systems, Time-Series Modelling, On-Device / Edge Inference

Achievements

- **'Tech for Good' Winner** — Ibec Technology Awards (2025)
- **Best New Sports Business Finalist** — Irish Sports Industry Awards (2024)
- **Commercialisation Award** — Dublin City University (2022)